

Special Senses

Eye & Ear

Clinical Integration

Clinical Case 1

A 48-year-old man comes to the physician because of a 1-month history of constant ringing in his ears. He is a combat veteran and recently returned from assignment. On examination, his hearing is grossly impaired. Acoustic testing shows sensorineural hearing impairment.

- a) List the components of cochlea canal.
- b) Vibration of which membrane ultimately leads to hearing?
- c) Explain how the loss of hair cells affect hearing.

Clinical Case 2

A 56-year-old man comes to the physician because of a 4-month history of failing vision. He has been diabetic for approximately 5 years, but has not been compliant with his medication, insulin, for the past 6 months. On examination, visual acuity is noticeably reduced in the right eye compared to the left. Fundoscopy shows the lens to be hazy bilaterally.

- a) State the origin of lens.
- b) State the name of the proteins found in the lens.
- c) Describe the changes to the collagen fibers of lens that leads to formation of cataract.

Clinical Case 3

A 53-year-old man comes to the emergency department with a 2-day history of severe headache, eye pain, red eyes, loss of vision, and vomiting. He reports seeing rings (halos) around the lights. Physical examination shows a reduction in the peripheral field of vision. Tonometry shows an elevated intraocular pressure which aided the diagnosis of acute angle closure glaucoma.

- a) State the site of production of aqueous humor.
- b) What type of capillaries are found in the ciliary processes?
- c) Explain the consequences due to blockage of canal of Schlemm.